

09/09/2026

ML FUNDAMENTALS + REGRESSION MODELS

1. Machine learning -

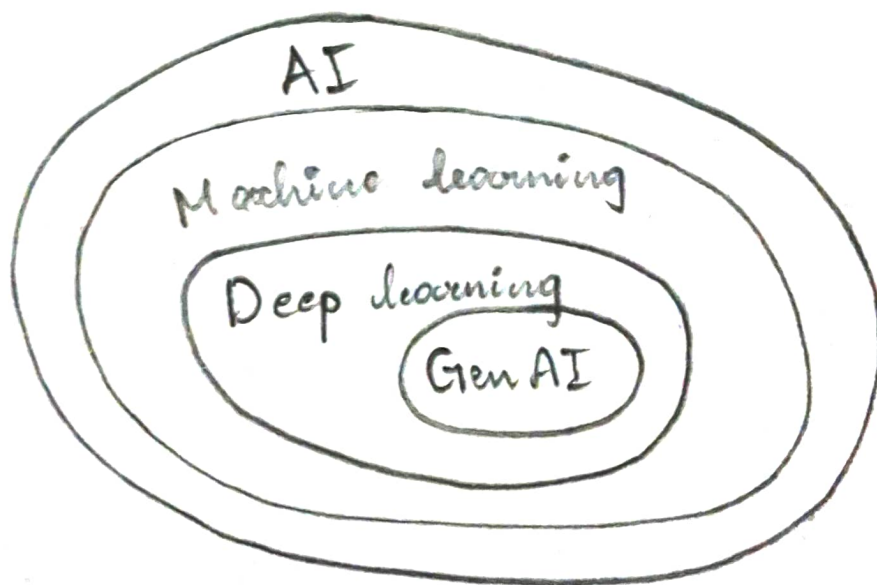
A branch of Artificial Intelligence where systems learn patterns from data and use them to make predictions or decisions without being explicitly programmed for every case.

2. Artificial Intelligence Vs ML Vs Deep learning

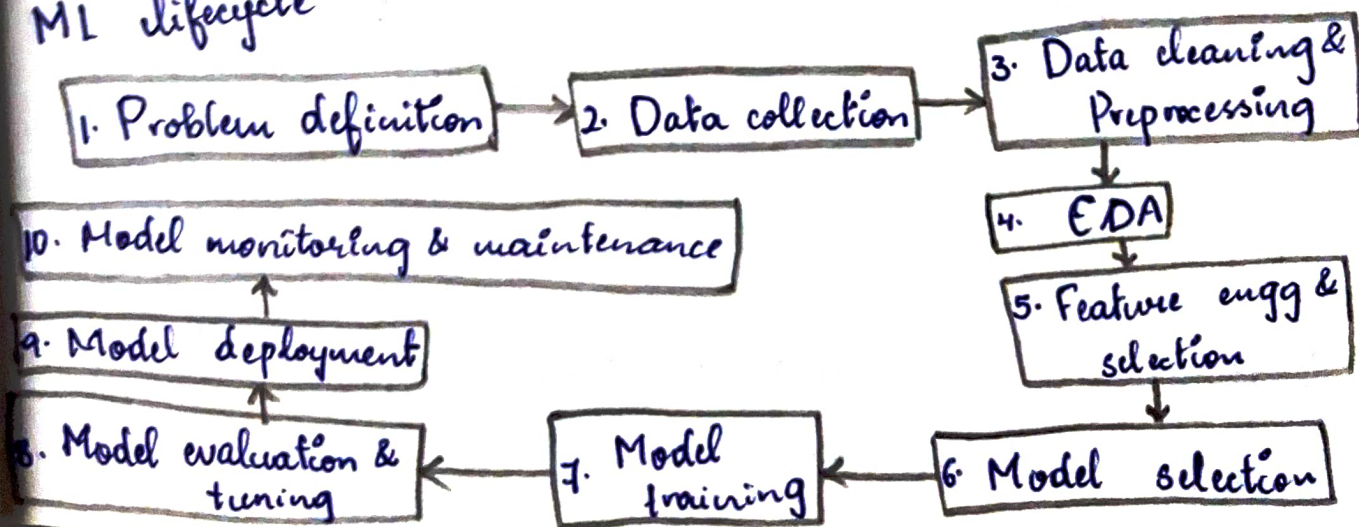
↓
broad field of making machines perform intelligent tasks

↓
machines learn patterns from data.

↓
Subset of ML that uses multi-layer neural networks



ML lifecycle



Applications -

1. Healthcare and medical
2. User experience
3. Autonomous vehicles
4. E-commerce
5. Agriculture
6. Smart assistants
7. Fraud detection
8. Cybersecurity
9. Personalized learning
10. Industrial automation

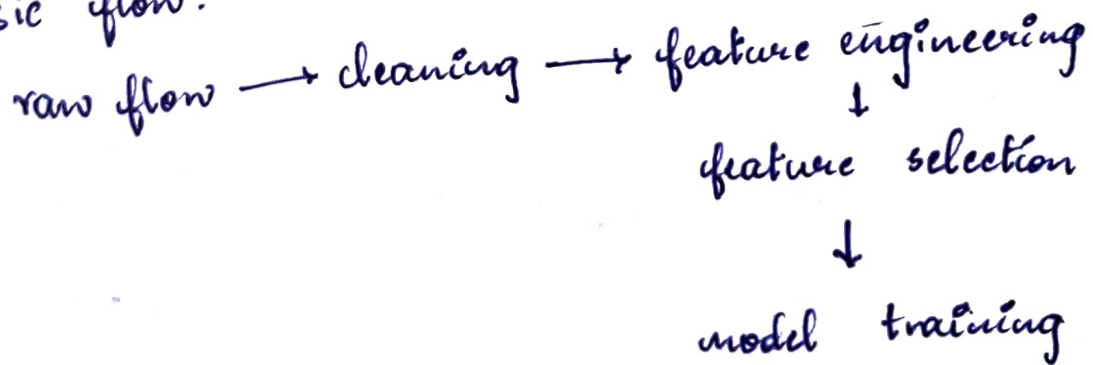
Types of Machine Learning

- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Reinforcement learning

3. Feature Engineering -

The process of creating, transforming, selecting or improving input features so that ML models can learn patterns from the data.

Basic flow:



1. Feature creation - creating useful new variables from existing data.

eg: DOB
↓
Age

price x quantity
↓
TAmount

ii. Feature Transformation - changing the representation or distribution of an existing feature.

example: log transformation

square root transformation

scaling

binning

iii. Feature Scaling - brings numerical features onto comparable scales.

o Standardization -

transforms data so that:

mean ≈ 0 , std ≈ 1

$$z = \frac{x - \mu}{\sigma}$$

mostly useful for -

KNN

SMM

logistic regression

PCA

neural networks.

o Normalization - rescale to range 0 - 1

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Encoding Categorical features -

→ ML models generally need numerical input.

• Label Encoding -

Assigns numbers to categories.

• One-hot encoding -

Creates a separate binary column for each category.

Red	Blue	Green
1	0	0
0	1	0
0	0	1

useful for nominal categories.

Handling missing values -

common methods:

- drop rows/cols
- mean imputation
- median "
- mode "

- forward/backward fill
- model-based imputation.

Handling Outliers -

- remove
- cap
- transform
- use robust statistics/models
- keep them if meaningful

Never remove an outlier automatically without understanding why it exists.

4. Feature Selection -

Choosing the most useful original features while removing redundant ones.

Benefits :

- overfitting ↓
- improves interpretability
- make models faster
- removes redundant info

Main Methods

- Filter methods - select features using statistical properties.

examples : Correlation, chi-square, ANOVA

- Wrapper Methods - evaluate diff features subsets using model performance.

example :

Recursive Feature Elimination - RFE

- Embedded Methods - feature selection happens during model training.

examples :

Lasso

Decision Tree feature importance

Random Forest " "

* Simple memory ↓

filter → before model

wrapping → repeatedly uses model

embedded → during model training

5. Dimensionality Reduction -

Reduces the number of input dimensions while trying to preserve important information

⇒ Difference :

1. Feature selection -

→ keeps some original features.

2. DR -

→ creates fewer transformed features.

• PCA (Principal Component Analysis) -

An unsupervised dimensionality reduction technique that transforms correlated features into fewer uncorrelated principal components while preserving as much variance as possible.

Conceptually -

many correlated features

↓

PCA

↓

fewer principal components

example :

100 features → PCA → 10 P.C

• principal components • variance • eigen values • eigen vectors

PC1 → captures max variance

PC2 → captures next-highest variance and is perpendicular to PC1.

• LDA (Linear Discriminant Analysis) -

A supervised dim-reduction technique that tries to maximize separation b/w known classes.

Main difference:

1. PCA -

→ unsupervised
→ preserves variance

2. LDA -

→ supervised
→ maximizes class separation

6. Multicollinearity -

occurs when two/more input features are strongly correlated with each other.

example: Age
Years of experience

possible problems -

- Coefficients become unstable
- Interpretation becomes difficult
- Redundant info is included

possible solutions -

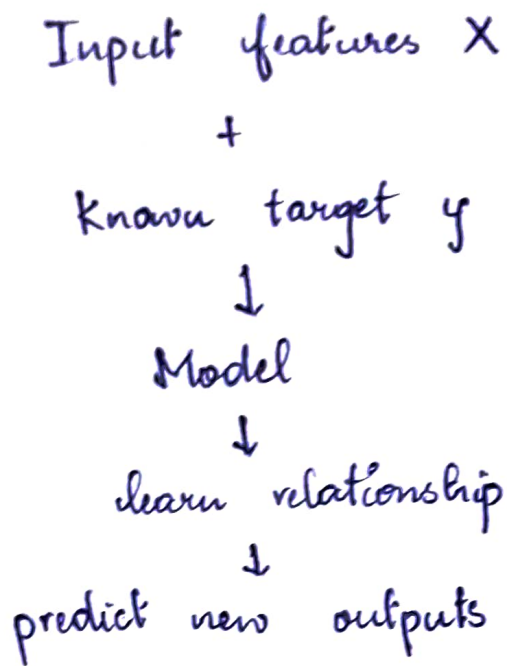
- remove correlated features
- Use PCA • use regularization
- use domain knowledge

7. Feature Importance -

Measures how strongly each feature contributes to a model's predictions.

8. Supervised learning -

A machine-learning approach where the model learns from labeled data containing both input features and the correct output.



Types: i. Regression ii. Classification

9. Regression - A supervised learning task where the target is a continuous numerical value.

examples: salary, house price

common models:

Linear Reg—

Decision Tree "

Random Forest "

Support Vector "

10. Linear Regression - A supervised regression algorithm that predicts a continuous value by fitting a linear

relationship between input features and the target

$$y = mx + b$$

input (independent) $\rightarrow$ x
predicted (dependent) $\leftarrow$ y
slope of the line $\leftarrow$ m
intercept $\leftarrow$ b

$\rightarrow$ multiple linear regression -

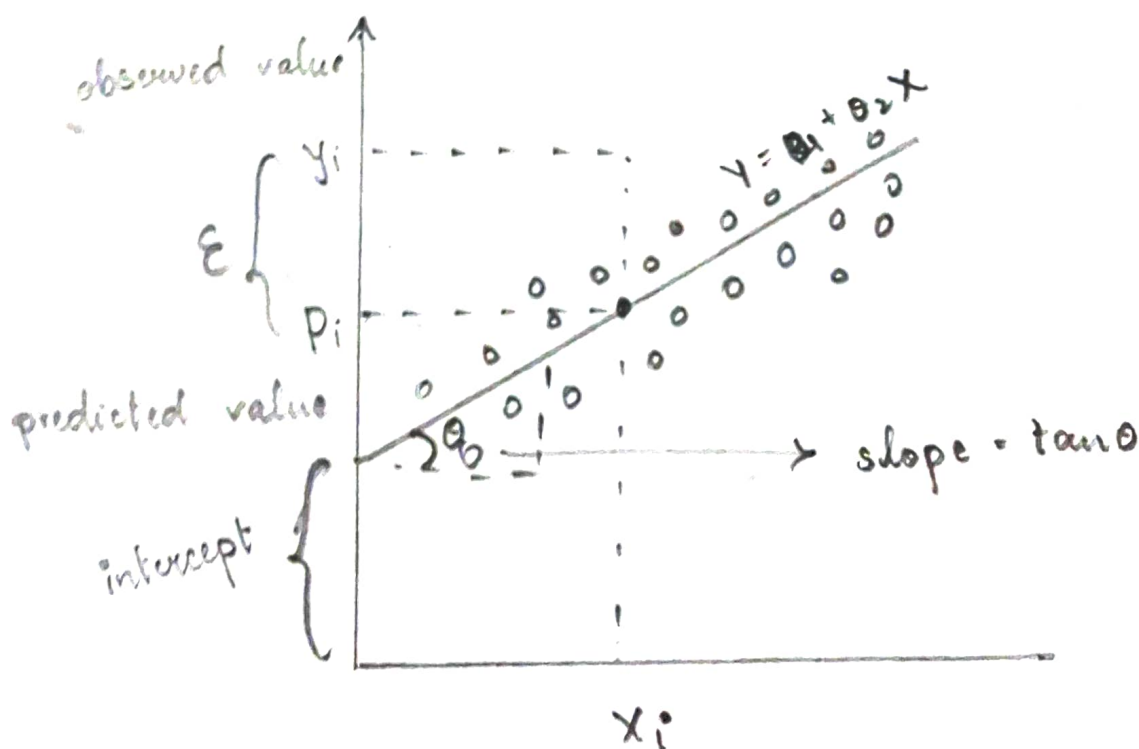
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

β_0 = y-intercept

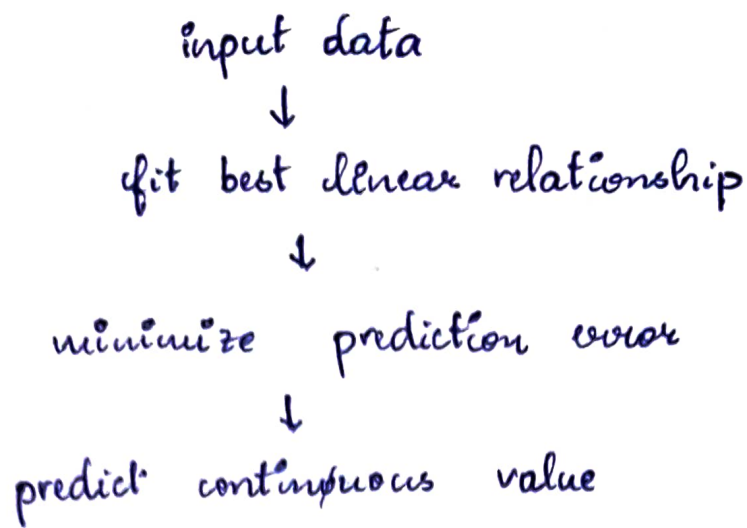
$\beta_1, \dots, \beta_k$ = slopes

$X_1, \dots, X_k$ = independent variables

ε = model error



* goal of best-fit line - find straight line that minimizes the error.



* coefficient - effect of a feature on prediction.

* intercept - predicted value when inputs are zero.

* residual - diff btw actual and predicted value

$$\text{residual} = y_i - \hat{y}_i$$

→ Least squares method — $\sum (y_i - \hat{y}_i)^2$

Assumptions -

- linear relationships
- independent observations
- constant error variance
- low multicollinearity
- residual normality

Advantages -

- simple
- fast
- easy to interpret
- good baseline model

Limitations -

- sensitive to outliers
- cannot capture complex nonlinear relationships
- multicollinearity affects coefficient interpretation

11. Classification -

A supervised learning task where the target belongs to a discrete class or category.

examples :

Spam / Not spam

Fraud / Not fraud

Disease / Healthy

Cat / Dog

Common classification models -

Logistic Regression

Decision Tree

Random Forest

KNN

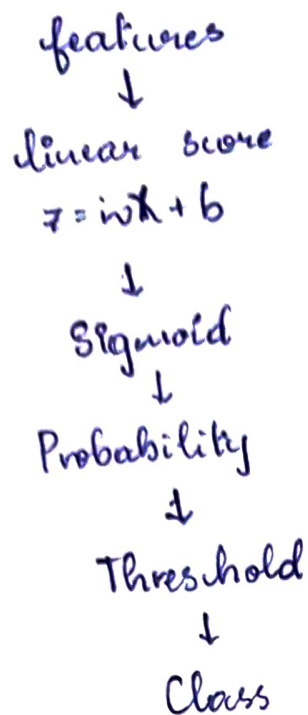
SVM

Naive Bayes

12. Logistic Regression -

A supervised classification algorithm that predicts the probability of an observation belonging to a class.

Working -

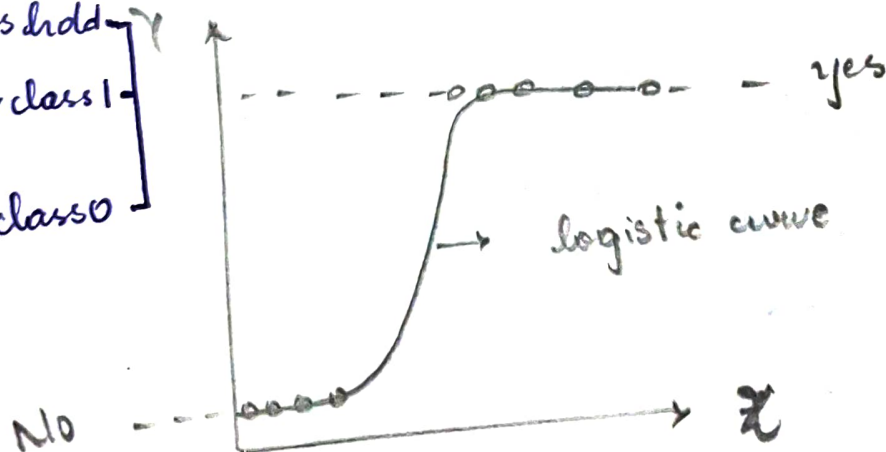


Sigmoid function - $\frac{1}{1+e^{-z}}$ o/p - 0 to 1

$$z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

↓
standard linear regression

Typical threshold
 $p \geq 0.5 \rightarrow \text{class 1}$
 $p < 0.5 \rightarrow \text{class 0}$



Advantages -

- Simple
- Fast
- gives probabilities
- easy to interpret
- good baseline classifier

Limitations -

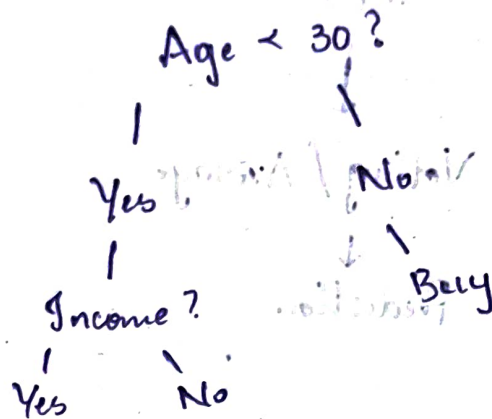
- Mainly models linear decision boundaries.
- Sensitive to extreme input values.
- Severe multicollinearity can cause instability.

13. Decision Tree -

A supervised learning algorithm that makes predictions by repeatedly splitting data according to feature-based conditions.

Used for:

- Classification
- Regression



- Root node
- Decision node
- leaf node
- Split
- Tree Depth

Classification splitting commonly uses:

- ↳ Gini Impurity
- ↳ Entropy
- ↳ Information Gain

Advantages -

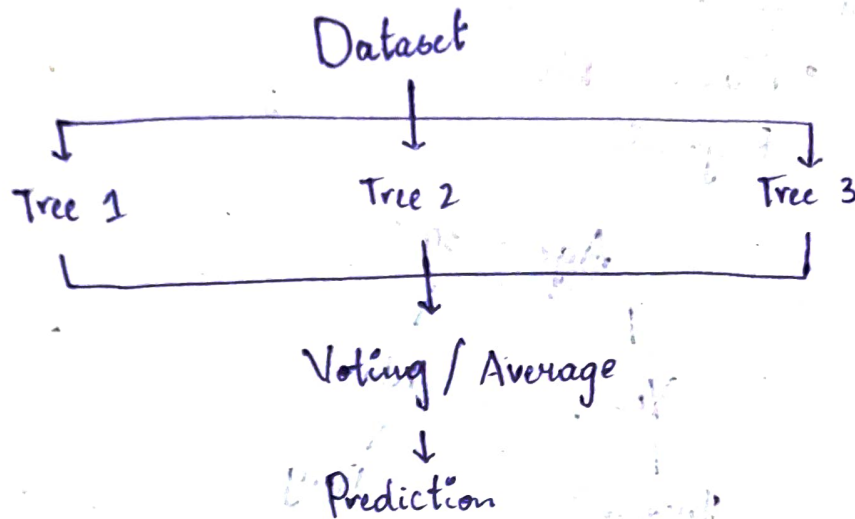
- Easy to understand
- handles nonlinear relationships
- little preprocessing required
- can handle numerical and categorical patterns

Limitations -

- easily overfits
- small changes in data can change the tree
- deep trees may generalize poorly

14. Random Forest -

An ensemble model that combines predictions from many decision trees to improve accuracy and reduce overfitting.



Uses :

Classification → majority voting

Regression → average prediction

Main Idea ↓

→ Random Forest introduces randomness through:

- Different data samples
- Diff — feature subsets

Advantages -

- Strong general-purpose model
- Handles nonlinear relationships
- Less overfitting than one decision tree.
- provides feature importance.

Limitations -

- Less interpretable than a single tree.
- Larger and slower than one decision tree.

15. K-Nearest Neighbors - KNN

A supervised learning algorithm that predicts based on the labels or values of the nearest training examples.

○ ○ ○ ○ ○
★ ← new point

The model finds the nearest (k) points.

→ For classification: Majority class wins
→ For regression: Average nearby values

KNN uses distance, commonly:

$$d = \sqrt{\sum (x_i - y_i)^2}$$

Advantages -

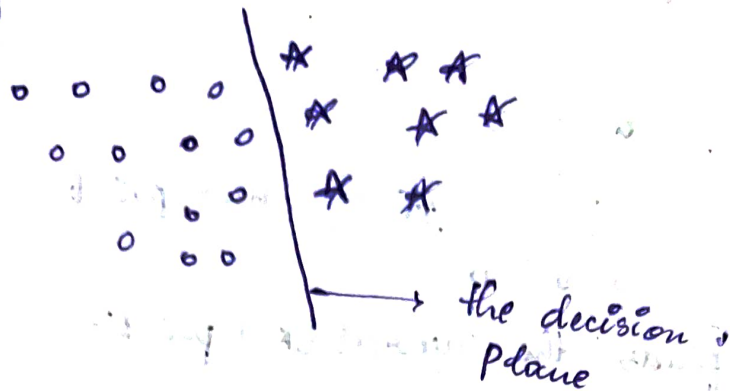
- Simple
- No traditional training phase
- Works well on smaller datasets

Limitations -

- Slow for large datasets
- Sensitive to feature scaling
- Sensitive to the value of k
- Performs poorly in very high dimensions.

16. Support Vector Machine -

A supervised algorithm that finds the decision boundary that maximizes the margin between classes.



The points closest to the boundary — support vectors

- Hyperplane - decision boundary ($w \cdot x + b = 0$)
- Support vectors -
- Margin - The distance b/w hyperplane and support vectors.
- Kernel — mathematical function that transforms low-D input data into higher-dimensional space to make non-linear data easily separable.
 - common kernels:
 - linear
 - polynomial
 - RBF

Advantages -

- Effective in high-dimensional space
- Works well on smaller & medium-sized datasets.
- Can model nonlinear boundaries using kernels.

Limitations -

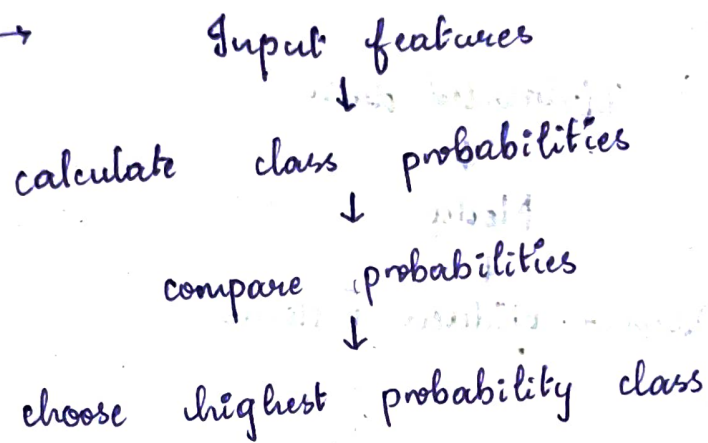
- Can become slow on very large datasets.
- Sensitive to parameter choices.
- Requires scaling.
- Harder to interpret.

17. Naive Bayes-

A probabilistic supervised classification algorithm based on Bayes' theorem, that assumes features are conditionally independent given the class.

$$P(\text{class}|\text{data}) = \frac{P(\text{data}|\text{class}) \times P(\text{class})}{P(\text{data})}$$

Working →



Why Naive?

- it assumes features are independent.
- often unrealistic assumption, but works surprisingly well.

Types :

- Gaussian Naive Bayes
- Multinomial "
- Bernoulli "

Typical uses :

- Spam detection
- Text classification
- Sentiment "

Advantages-

- Very fast
- Works well with text data
- Handles high-dimensional data well
- Requires relatively little training data

Limitations-

- Independence assumption is often unrealistic.
- Performance may suffer when features are strongly dependent.